EX NO: 1

Date : 23/07/26

PRACTICAL - 1 BASIC NETWORKING COMMANDS

AIM:

To study of various Network commands used in Linux and Windows.

1. arp - a :

ARP is a short form of address resoultion protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

OUTPUT:

Address	HW Address	HW Type	Flags Mask	Iface
(172.16.10.116) at	d8:bb:c1:c5:c8:c6	ether	C	enp3s
172.16.8.188	7C:57:58:38:45:3b	ether	C	enp3s
172.16.11.74	00:27:0e:13:ef:04	ether	C	enp3s
172.16.10.110	d8:bb:c1:c5:ca:0a	ether	C	enp3s
172.6.8.150	00:27:02:13:2d:88	ether	C	enp3s

2. host name :

This is the simplest of all TCP/IP commands. It simply displays the name of your computer

OUTPUT:

localhost. local domain

3. ipconfig/all :

This command displays detailed configuration information about your TCP/IP connection including router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system

OUTPUT:

Windows IP configuration

Ethernet adapter Ethernet :

Connection - Specific DNS Suffix . :

Link - local IPv6 Address . . . . . : fe80::70a6:94b3:c428:4e0%14

IPv4 Address : 172.16.10.78

Subnet Mask . . . : 255.255.252.0

Default Gateway . . . : 172.16.8.1

4. nbtstat -a:

This command helps solve problems with NetBIOS name resolution (Nbt stands for NetBIOS over TCP/IP)

OUTPUT:

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP)

: NBTSTAT [[-a Remote Name] [-A IP address] [-c] [-n]
[-r] [-R] [-RR] [-s] [-S] [interval]

Remote Name	Remote host machine name
IP address	Dotted decimal representation of IP address
Interval	Redisplays selected statistics, pausing interval seconds

5. netstat:

(network statistics) netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc

Eg: netstat -r

OUTPUT:

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	localhost.localdoma:ssh	172.16.10.87 :49985	Established
tcp	0	0	localhost.localdoma:ssh	172.16.10.88 :60935	Established
tcp	0	0	localhost.localdoma:ssh	172.16.8.174 :52940	Established
tcp	0	0	localhost.localdoma:ssh	172.16.8.130 :21548	Established
tcp	0	0	localhost.localdoma:ssh	172.16.9.228 :5943	Established

6. nslookup:

(name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in two modes: interactive and non-interactive.

Eg: nslookup www.googl.com

OUTPUT:

Server: 172.16.8.1

Address: 172.16.8.1#53

Non-authoritative answer:

Name: www.googl.com

Address: 142.251.153.119.

7. Pathping:

Pathping is unique to windows and basically a combination of the ping and Tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering, statistics on the rate of data loss along each hop

OUTPUT:

Tracing route to www.googl.com [142.251.154.119]

Over a maximum of 30 hops:

0 DESKTOP-96Q396B [172.16.10.78]

1 172.16.8.1

2 137.97.163.129

3 * * *

www.googl.com ping statistics

44 packets transmitted, 44 recieved, 0% packet loss, time 42978ms

rtt min/avg/max/mdev = 2.752/8.147/20.472/4.570 ms

8. PING google.com:

Check whether another computer on website is reachable over the network using ICMP Echo Request and Reply messages

OUTPUT:

Pinging google.com [142.251.221.174] with 32 bytes of data:
Reply from 142.251.221.221.174: bytes=32 time=12ms TTL=120
Reply from 142.251.221.174: bytes=32 time=12ms TTL=120

Ping statics for 142.251.221.174:
Packets: Sent=4, Received=2, Lost=2 (50% loss)
Approximate round trip time in ms:
Minimum=12ms, Maximum=12ms, Avg=12ms.

9. Route:

Displays or modifies the IP routing table used by the operating system

OUTPUT:

Manipulates network routing tables
ROUTE [-f] [-p] [-4] [-6] command [destination] [mask netmask]
[gateway] [Metric metric] [IF interface]

LINUX COMMANDS

10. ip address show:

Displays and manages IP addresses, routing and network interfaces.

OUTPUT:

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536
qdisc noqueue state UNKNOWN group default link/loopback
00:00:00:00:00:00 brd
00:00:00:00:00:00
inet 127.0.0.118 scope host lo
valid_lft forever preferred_lft forever

11. ifconfig:

Displays network interface details such as IP address and network statistics.

OUTPUT:

etho: flags = 4163 < UP, BROADCAST, RUNNING, MULTICAST > mtu 1500

inet 172.17.197.94 netmask 285.255

inet 6 fe80::215:5dff:fe82:dc86

Prefixlen 64 Scopeid 0x20 <link>

12. mtr google.com:

Combines ping and Traceroute to display the network path and performance statistics

OUTPUT:

DESKTOP-JSUHOT4A (172.17.197.94) → google.com

HOST	Loss %	Snt	Last
1. DESKTOP-JSUHOT4A (172.17.197.94) → google.com	0.0%	128	0.1
2. 192.168.1.1	0.0%	146	4.7
3. 100.84.0.1	0.0%	163	5.7
4. waiting for reply			

13. tcpdump -i etho :-

Captures and displays network packets passing through a network interface for analysis

OUTPUT:

tcpdump: verbose output suppressed, use -v or -w for full protocol decode

listening on etho link-type ENIOMB (ethernet), captures size 262144 bytes

10:15:21:123456 IP 192.168.1.10.52134 > 8.8.8.8.53:

UDP, length 32.

14. Ping google.com :

Sends ICMP ECHO request packets to verify network connectivity and measure response time

OUTPUT :

PING google.com (142.251.43.206) 56(84) bytes
64 bytes from icmaaa-bb.in-f14.1e100.net
(142.251.43.206): icmp-seq=1 ttl=117
time = 5.76ms

REF

RESULT:

Thus the various windows and linux networking commands were executed successfully